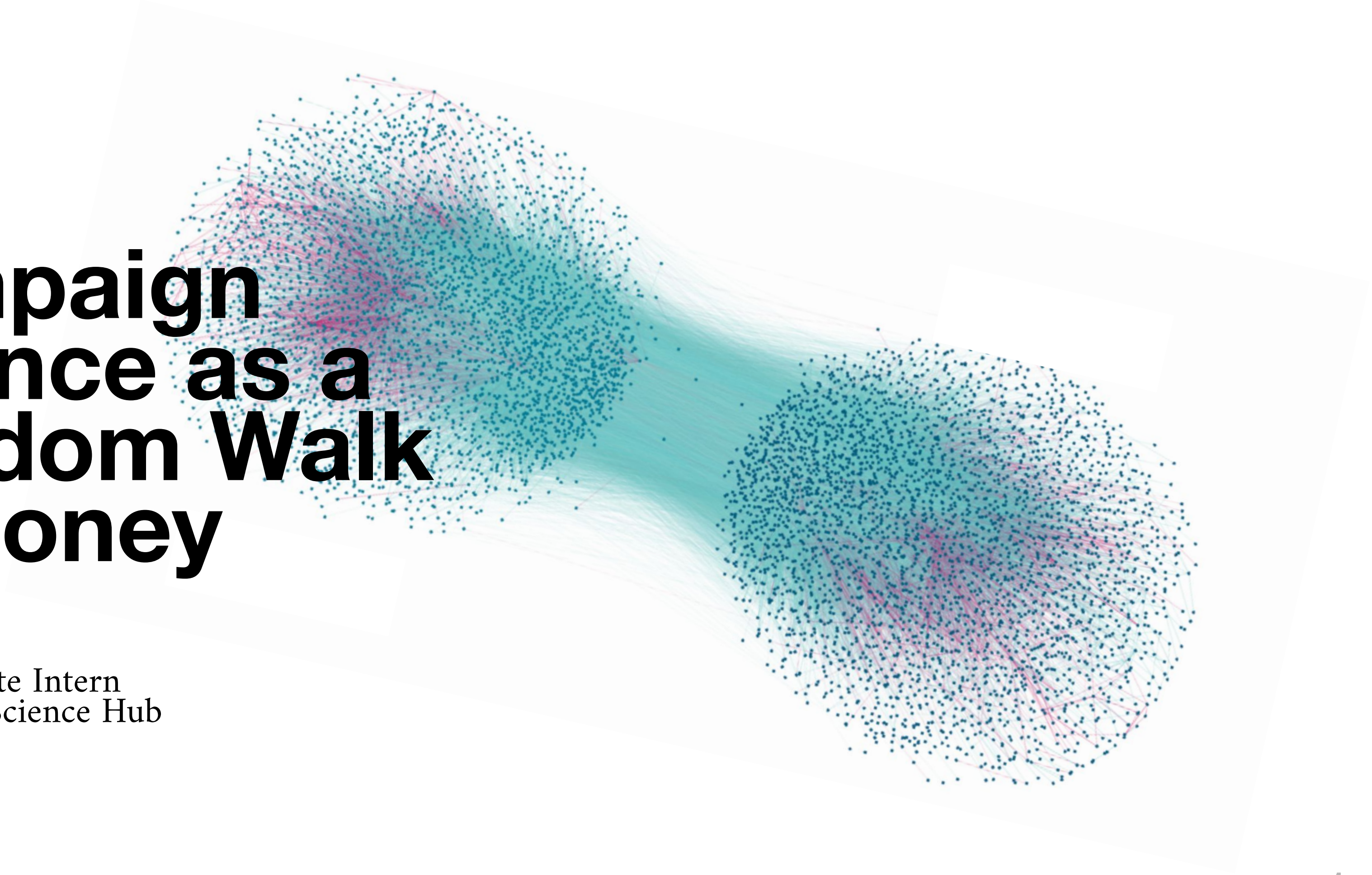
Campaign Finance as a Random Walk of Money



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Complexity Science Hub
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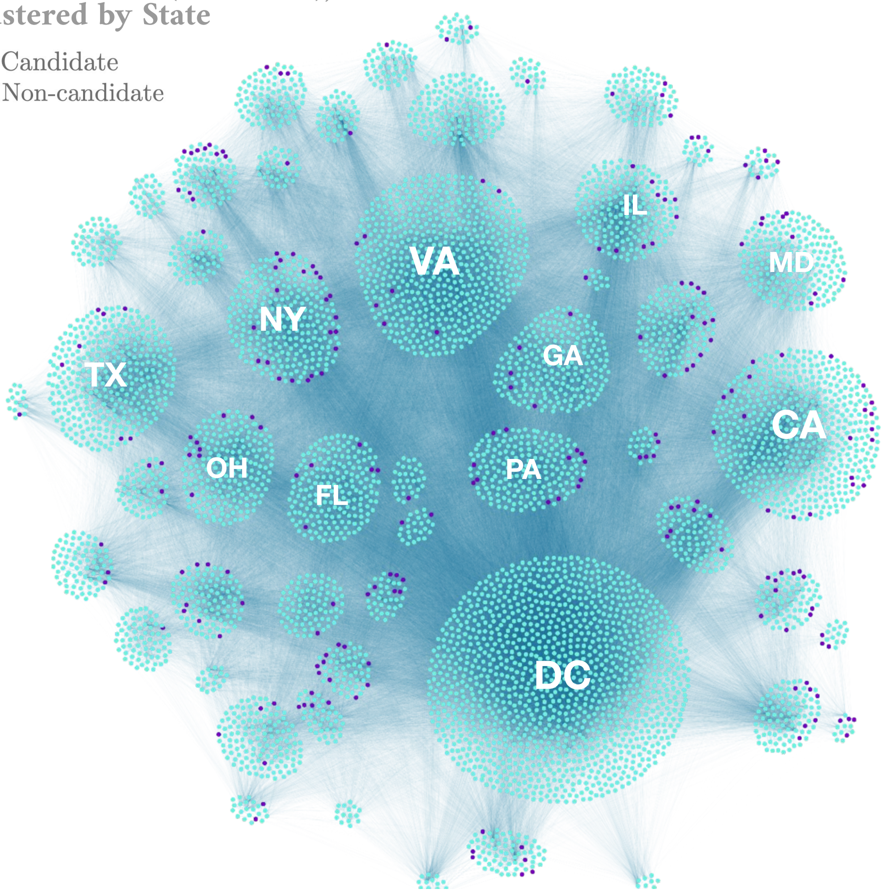
The (not so) hidden world of campaign finance

How does money move?
Does it differ by party?
What does it reveal about the network?

2022 Network (Midterms), Clustered by State

Candidate
Non-candidate



U.S. Federal Election Commission (FEC) Data, 2000–2022

Billions of dollars in transactions, publicly available.

Even-numbered years: Elect legislature every 2 years, president every 4 years.

Individuals excluded (no ID).

Keep only explicit entity-to-entity donations. Exclude loans, refunds, expenditures, memos.

Most entities' parties are unknown.

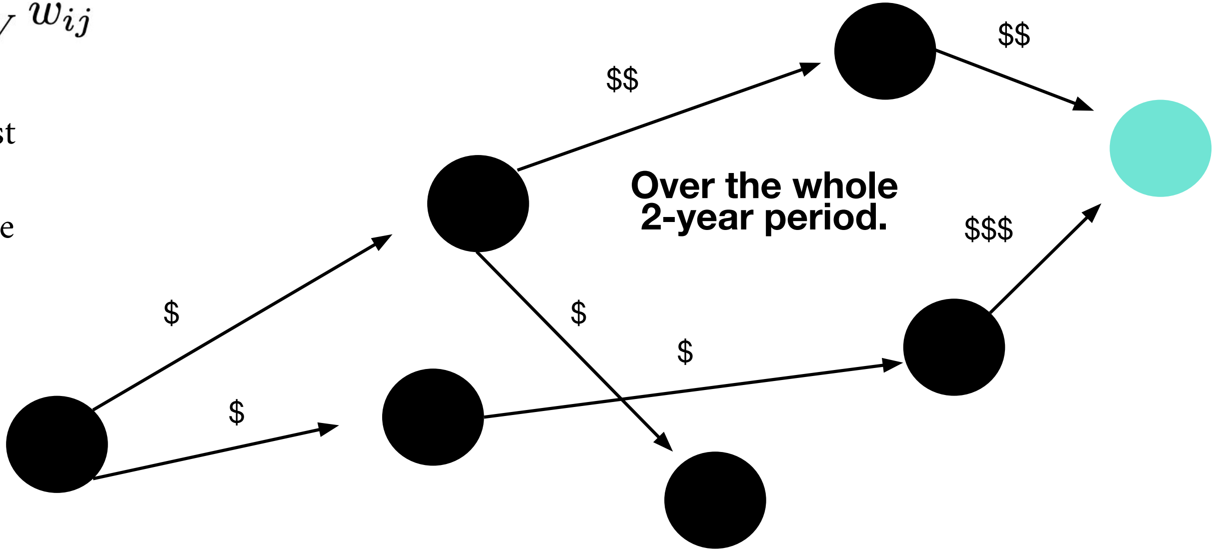
	Literature	My Project
1. Classify party	Spatial models (*a priori* homophily), parameter estimation (Bonica 2014)	**Simple absorbing Markov chain framework**: • Ideology arises purely from network properties • Absorption probabilities and absorption times
2. Partisan analysis	Traditional network metrics like centrality and clustering	

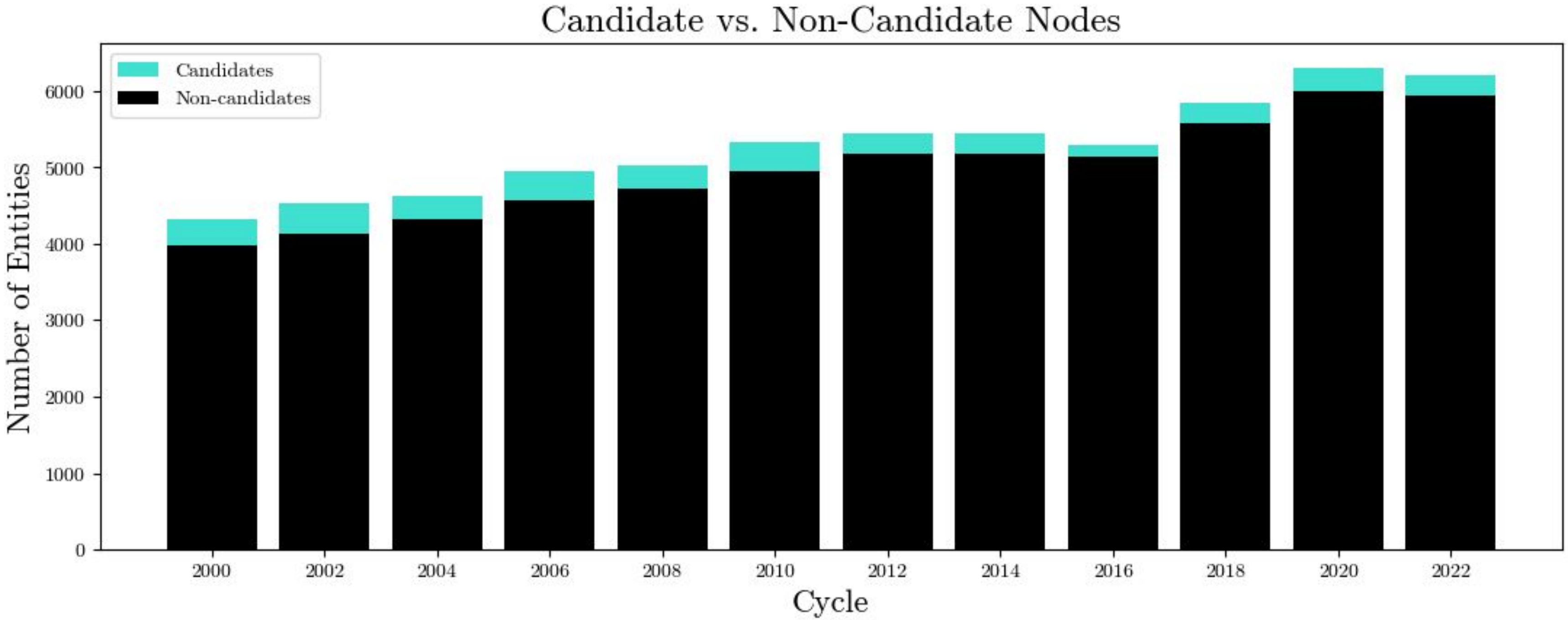
1. Model: An Absorbing Markov Chain

$$\mathbb{P}\{i \to j\} = \frac{w_{ij}}{\sum_{j \in V} w_{ij}}$$

(1) There exists at least 1 absorbing state;

(2) There exists a finite path from every state to the absorbing set.




Candidate vs. Non-Candidate Nodes
Candidates
Non-candidates
Number of Entities
0
1000
2000
3000
4000
5000
6000
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
Cycle

2. Label Propagation: Individual Ideology…

"I am the weighted average of my neighbors"

$$s_D(i) := \frac{\sum_{j \in V} s_D(j) w_{ij}}{\sum_{j \in V} w_{ij}}$$

$$s_R(i) := \frac{\sum_{j \in V} s_R(j) w_{ij}}{\sum_{j \in V} w_{ij}}$$

$\mathbf{S}^{(t+1)} = \mathbf{T}\mathbf{S}^{(t)}$ Iterate until convergence using the Markov transition matrix.

…Emerges from Global Absorption Probabilities

Analytic solution

$$\mathbf{S}_U = \mathbf{B}\mathbf{S}_L$$

Unlabeled nodes scores

Absorption probability matrix

Labeled node scores

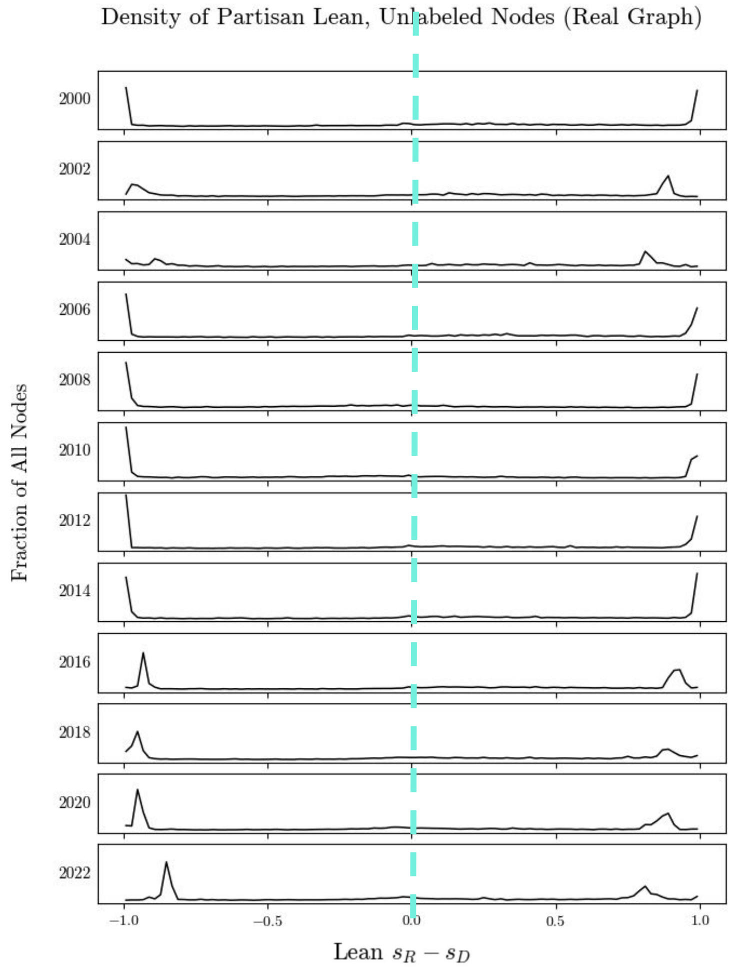
A normalized measure

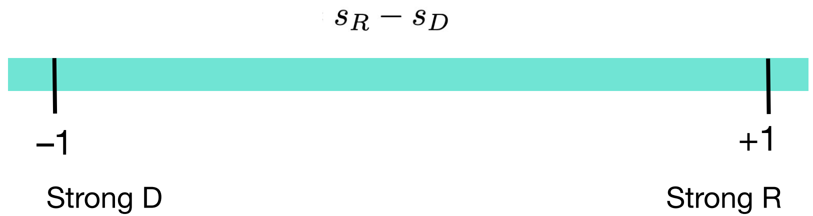
$$s_D(i) \equiv \mathbb{P}\{i \rightsquigarrow L \cap D\}$$

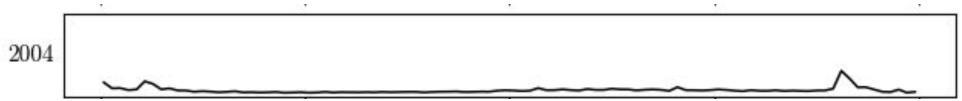
$$s_R(i) \equiv \mathbb{P}\{i \rightsquigarrow L \cap R\}$$

$$\therefore \quad s_D(i) + s_R(i) = 1$$

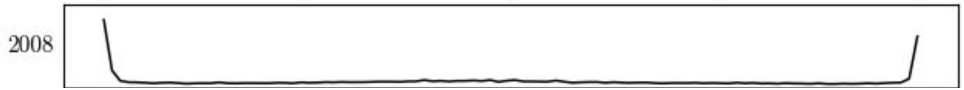
Dual: The score is a **harmonic extension** diffusing ideology from boundary to interior







President Bush (R, incumbent): Relied on swing state Ohio



Obama (D, first African-American)

The results:
Very clear bi-polarization, which motivates a binarized analysis.

Ideology initialized only on a few boundary nodes diffuses into stark basins.

Three Null Hypotheses

Unweighted (Remove Weights)

Observed effects are purely due to connectivity.

Permuted Weights (Global Shuffle)

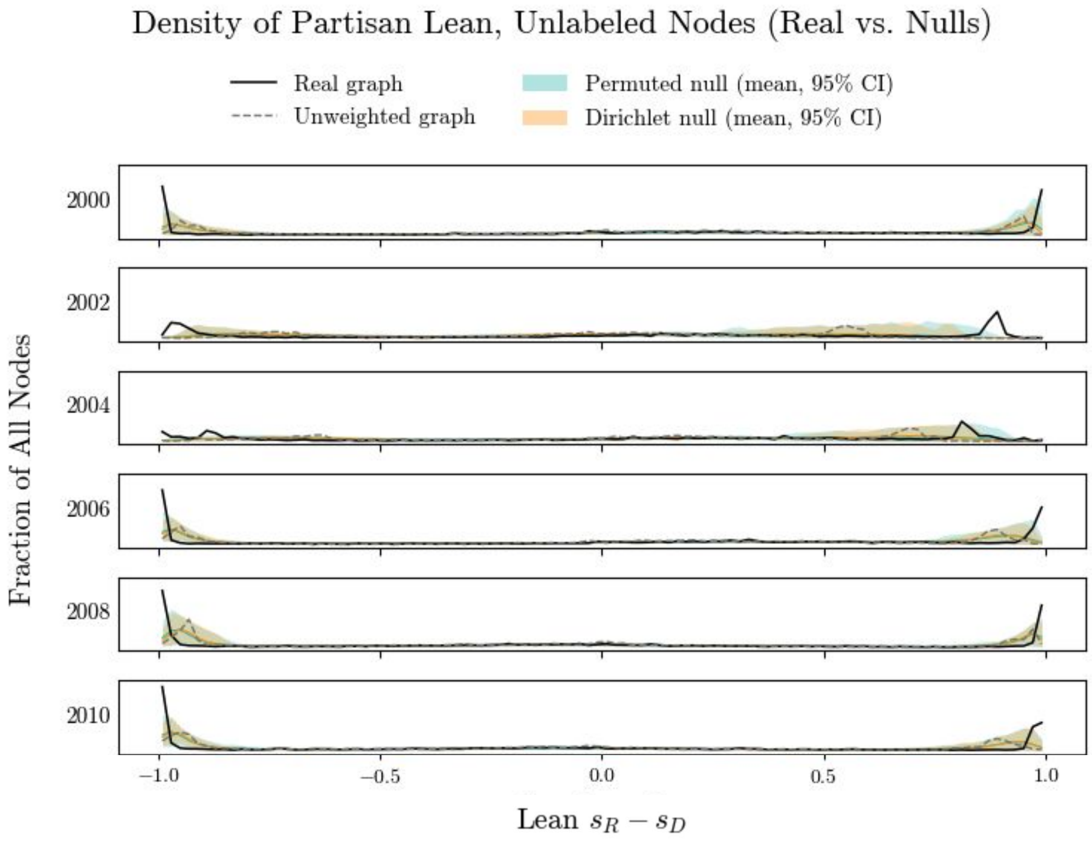
Observed effects do not rely on the precise locations of weights.

Dirichlet Weights (Individual Shuffle)

Observed effects do not rely on individual allocations of money.

The scores are highly polarized due to global and local weight distributions.

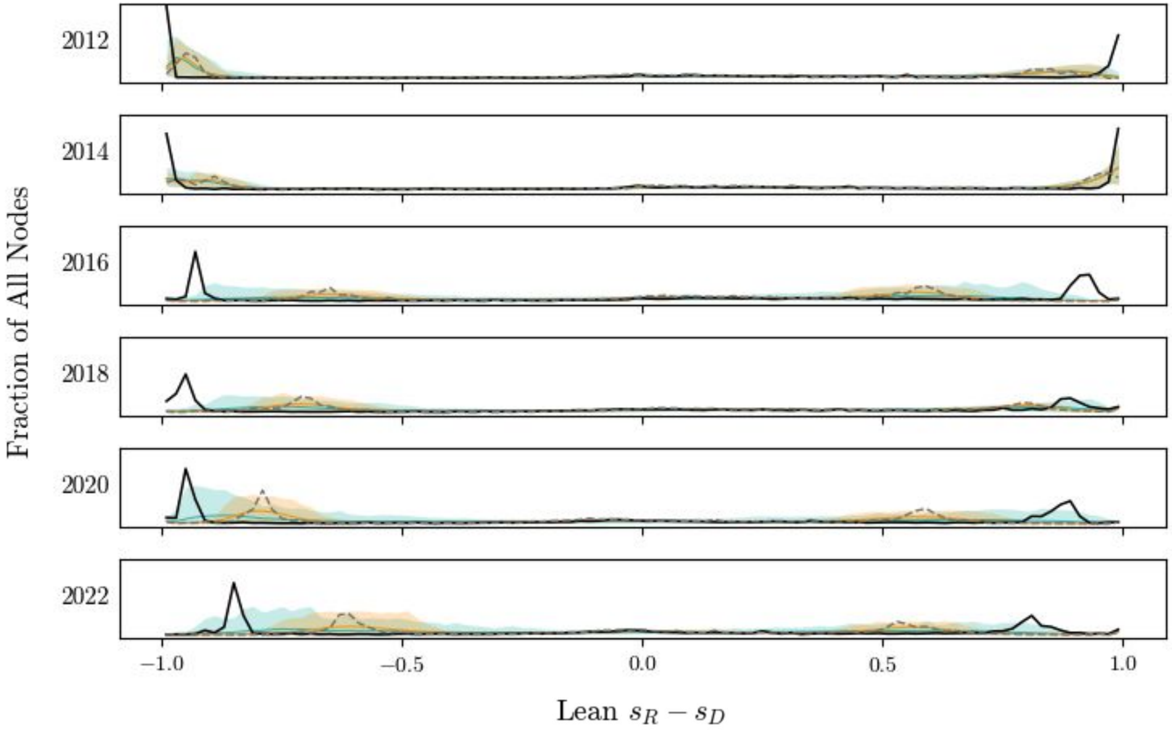
Topology can establish a broad structure…



… but it becomes *increasingly* more relevant **where** the money flows and **how** individuals distribute them.

Flow can reveal more than connectivity!

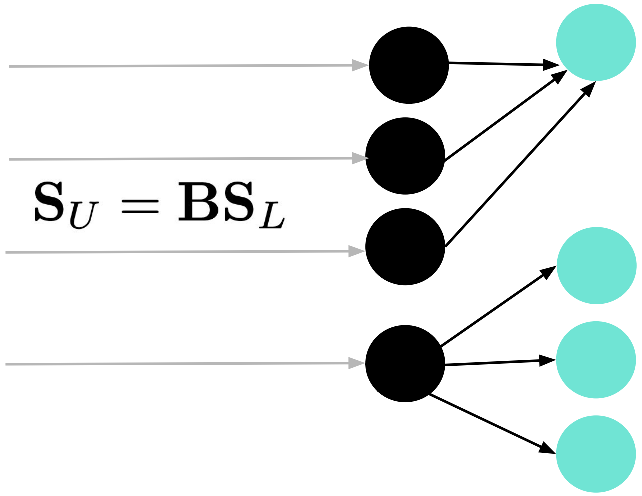
But what are the partisan Markov dynamics underlying this?



3. In-Party Financial Dynamics: Absorption Probabilities

How **spread out among candidates are absorbed dollars** that began with Democrats, versus with Republicans?

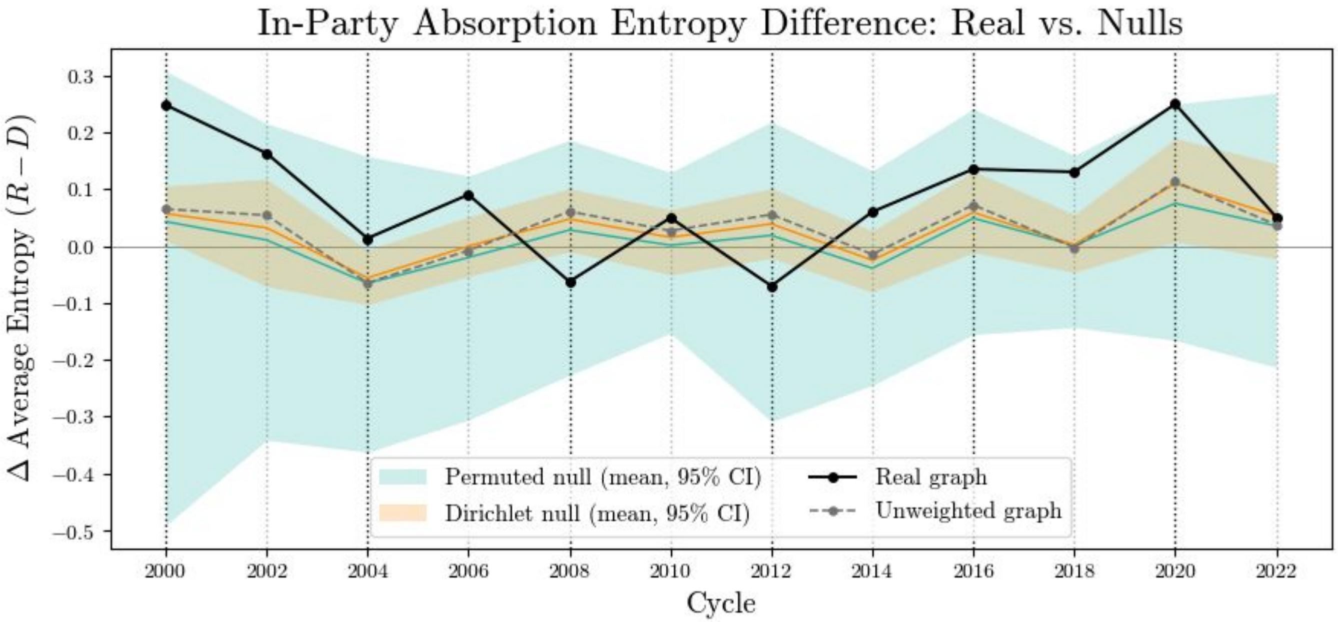
I compare the cycle-normalized **entropy of the absorption probability distributions** of Democratic and Republican transient nodes.



$$\mathbf{S}_U = \mathbf{B}\mathbf{S}_L$$

Republicans have slightly less focused absorption distributions.

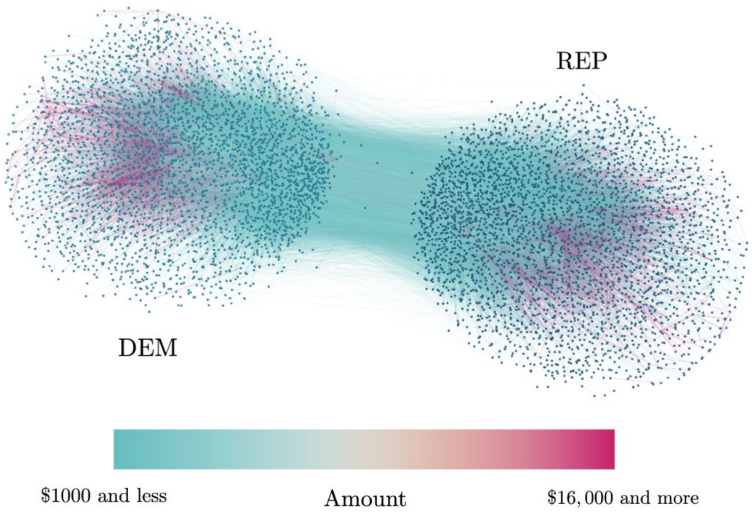
Since weight randomizations preserve some effect, it is attributed to topology: Nodes surrounding Republican candidates are more connected.



3. Cross-Party Financial Dynamics: Absorption Probabilities

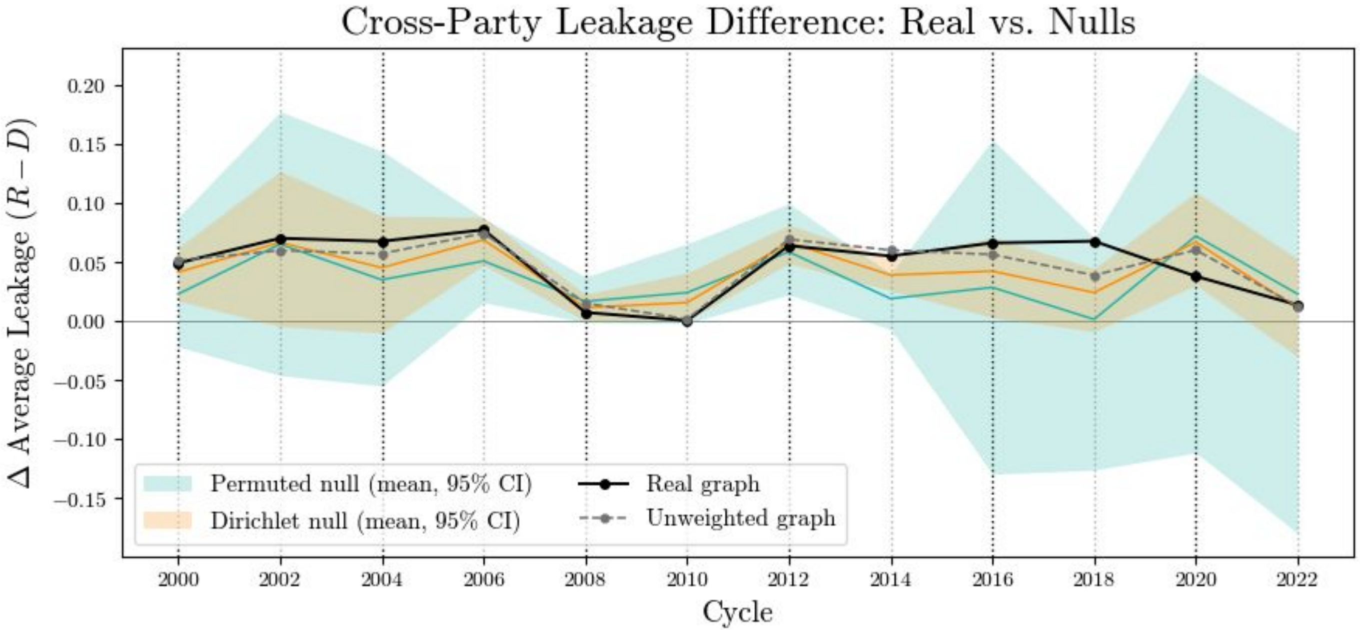
How much does money **cross the ideological barrier?**

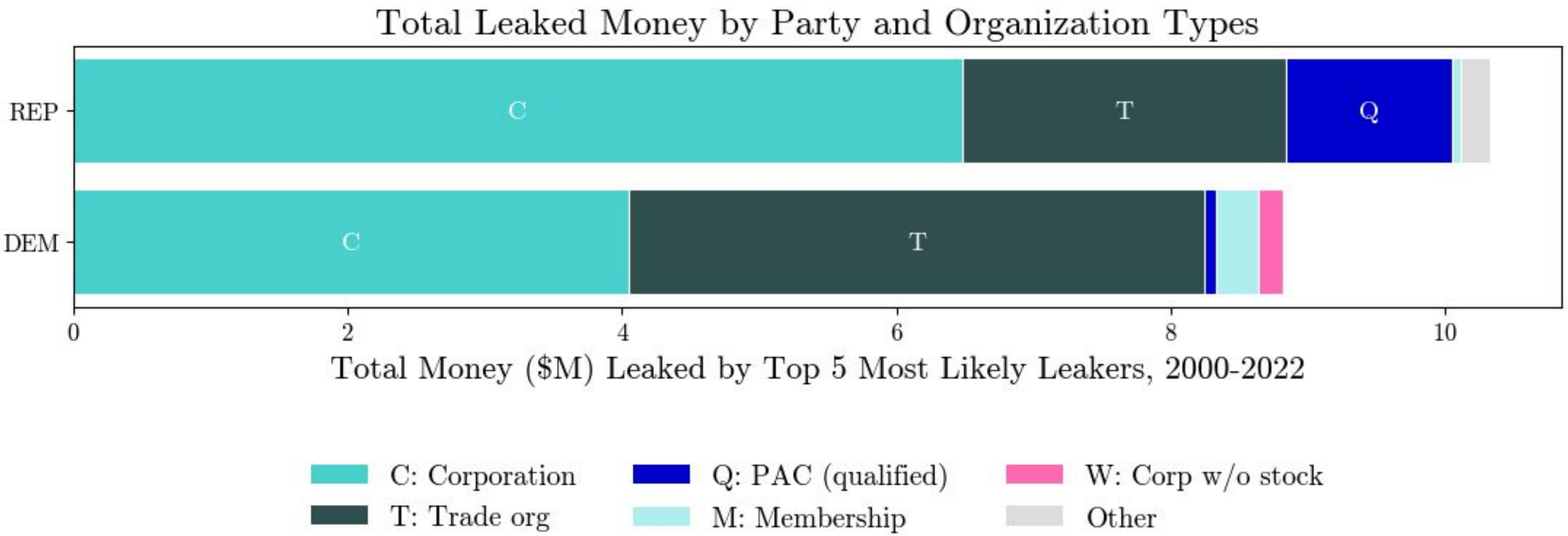
I compare the **absorption probabilities of dollars ending in the opposite party** between Democratic and Republican transient nodes.



Republicans have more leakage.

Since weight randomizations preserve some effect, it is attributed to topology: Party bridges are more likely to begin in right-leaning nodes.

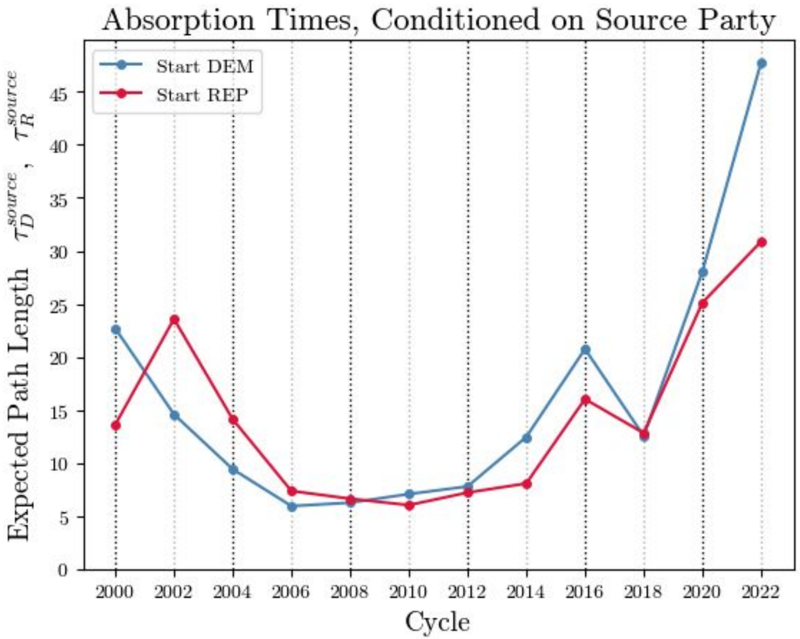


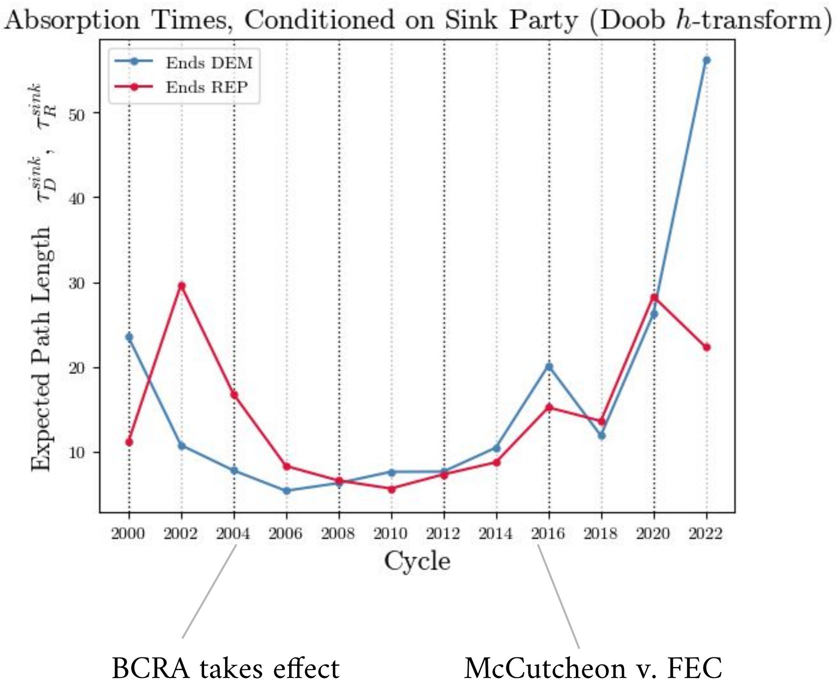


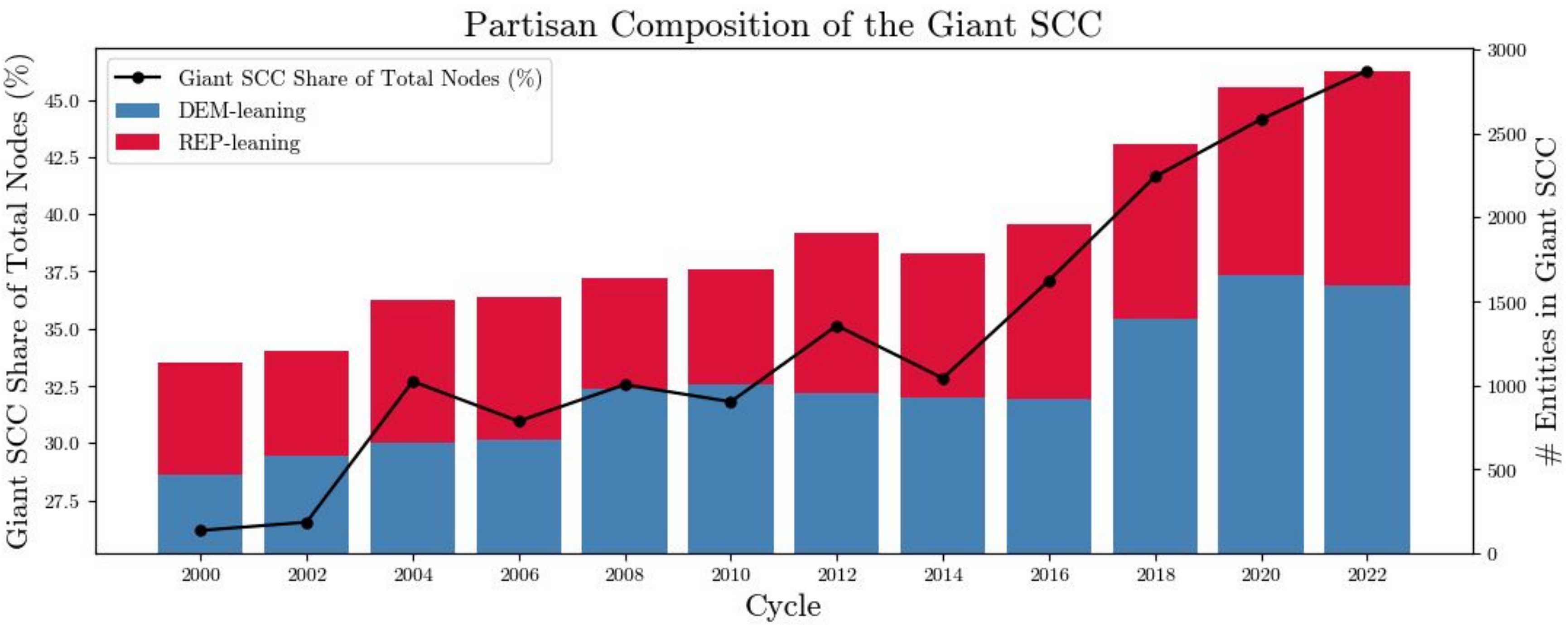
Those bridges begin in corporations and trade organizations.
Both in- and cross-party absorption shows that Republican-leaners have more diverse investment strategies.

4. Complexity: Absorption Times

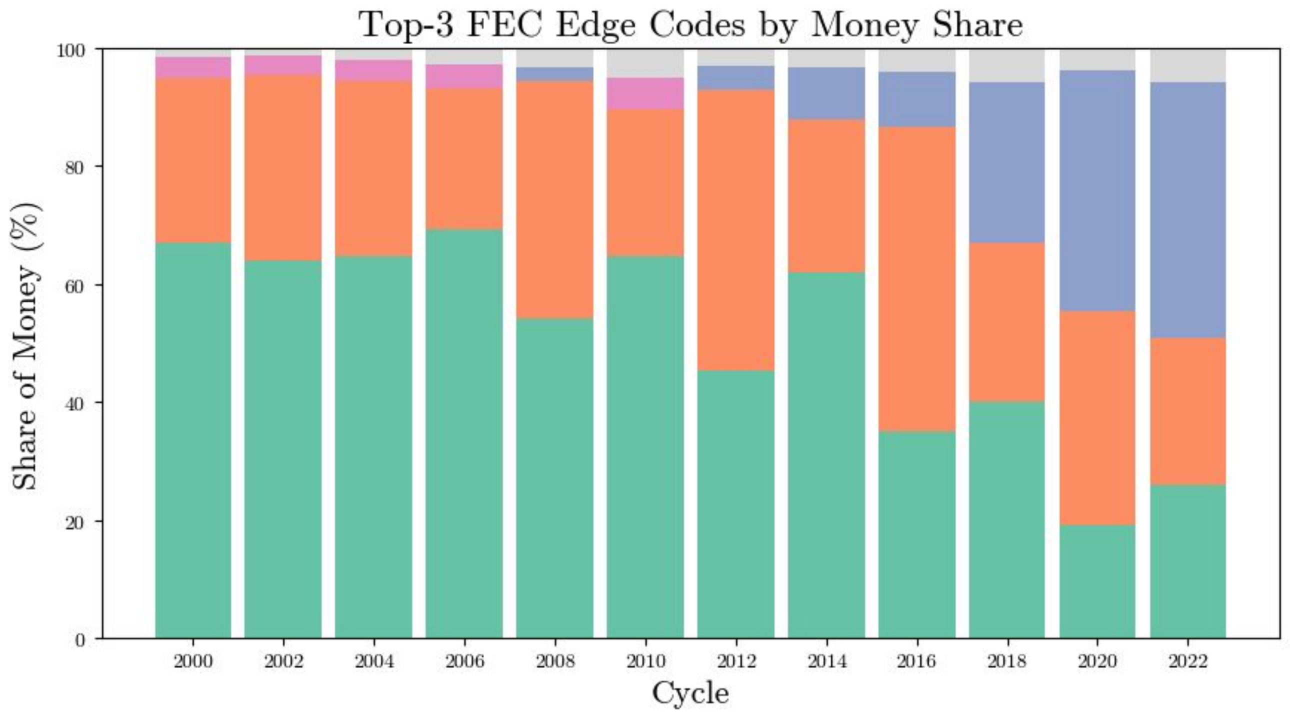
Path lengths are similar for both parties.

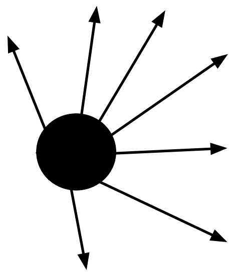






The network as a whole is becoming **increasingly well-connected**, causing large **basins where money can re-circulate and lengthen absorption times.**





A noticeably growing dominance of **24T conduit transactions**:
ActBlue (2004), WinRed (2019) consistently emerge as top 2 actors in recent years.

Conclusions

A novel absorbing chain framework for this problem shows that…

- "Individual" ideology is an emergent property of the random walk of dollars through the whole network.
- Even very little boundary conditions result in a **highly bi-polar ideological landscape**, which motivates a bipartisan analysis.
- **Republican and Democratic money follow equally complex trajectories** (absorption times) but **Republican money ends in a greater variety of destinations** (absorption probabilities). Republican-leaning actors have more diversified investment strategies, whereas Democratic-leaning actors spearhead a smaller selection.